

Lab 4: Data Structures and Factors

Level: Foundation Mapping: LO2 | K3 | A5 | A9 Concepts: matrix, data frame, list, factor, subsetting

Scenario

A fleet analyst needs the correct R structure for mixed vehicle records.

Goal

A typed data frame, a matrix slice and a factor frequency table.

Files in this folder

- lab-04.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Import the vehicle table from Excel.
2. Create a numeric matrix for selected measures and extract the specified slice.
3. Create a data frame with numeric and categorical columns.
4. Convert transmission to an ordered factor and produce a frequency table.
5. Bundle data and summaries in a named list.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-04.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-04.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Matrix contains only numeric values and has the expected dimensions.
- [] Data frame preserves column types.
- [] Factor levels and counts are explicit and correct.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.